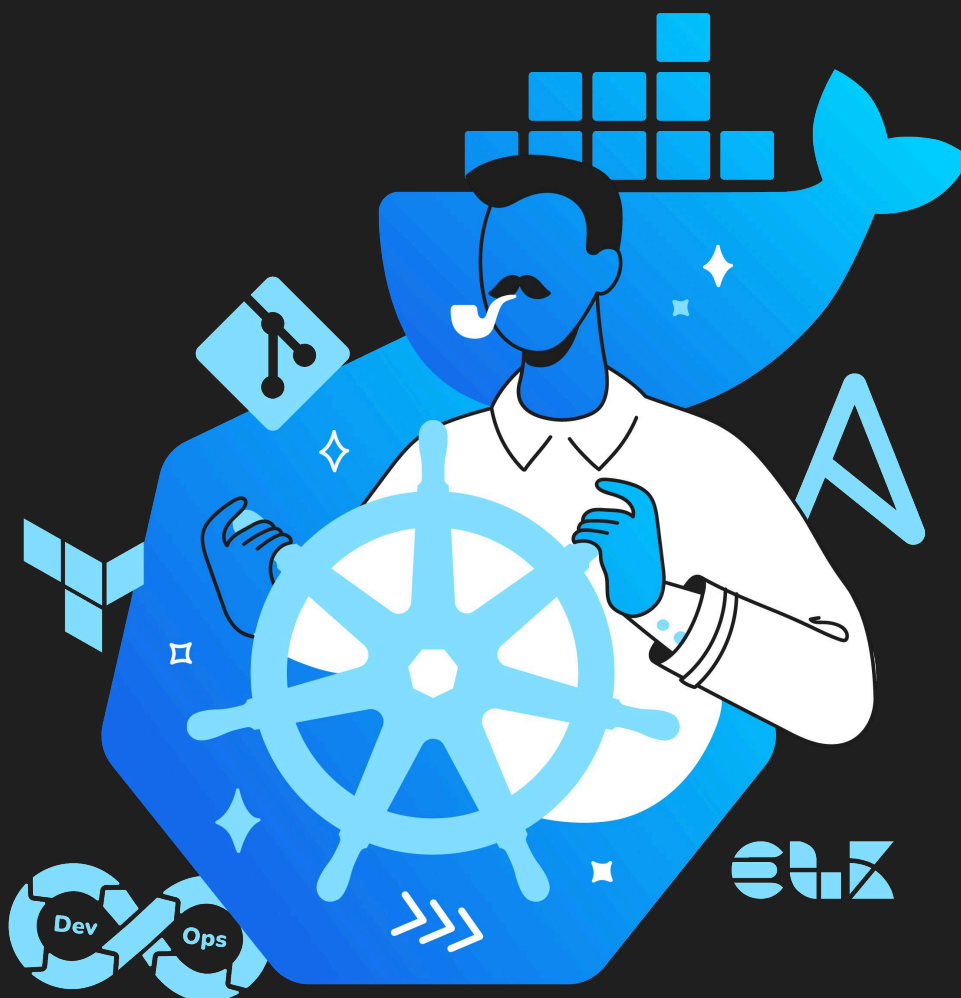


Методический материал

ОПТИМИЗАЦИЯ ОБРАЗОВ DOCKER

Урок 6.





Проводить работы по оптимизации мы будем с использованием 3-х Dockerfiles (Dockerfile-python-1, Dockerfile-python-alpine-2, Dockerfile-python-alpine-3).

Содержимое файла Dockerfile-python-1:

```
FROM python
```

Содержимое файла Dockerfile-python-alpine-2:

```
FROM alpine
RUN apk add --update-cache
RUN apk add python3
RUN apk add python3-dev
RUN apk add py-pip
RUN apk add build-base
RUN rm -rf /var/cache/apk/*
```

Содержимое файла Dockerfile-python-alpine-3:

```
FROM alpine
RUN apk add --update-cache \
    python3 \
    python3-dev \
    py-pip \
    build-base \
    && rm -rf /var/cache/apk/*
```



Переходим в терминал, перемещаемся в директорию с сохраненными докерфайлами:

```
$ ll
```

```
filipp@user-IdeaPad-3-15IAU7:~/dockerfiles$ ll
total 20
drwxrwxr-x  2 filipp filipp 4096 апр 14 13:16 ./
drwxr-x--- 31 filipp filipp 4096 апр 14 13:16 ../
-rw-rw-r--  1 filipp filipp   14 янв  1 1980 Dockerfile-python-1
-rw-rw-r--  1 filipp filipp  163 апр 14 13:14 Dockerfile-python-alpine-2
-rw-rw-r--  1 filipp filipp  132 апр 14 13:14 Dockerfile-python-alpine-3
```

Запускаем сборку образа на основе Dockerfile-python-1:

```
$ docker build . -f Dockerfile-python-1 -t docker-python-1
```

```
filipp@user-IdeaPad-3-15IAU7:~/dockerfiles$ docker build . -f Dockerfile-python-1 -t docker-python-1
[+] Building 38.6s (5/5) FINISHED
=> [internal] load build definition from Dockerfile-python-1                                docker:default
=> => transferring dockerfile: 60B                                                         0.0s
=> [internal] load metadata for docker.io/library/python:latest                         3.1s
=> [internal] load .dockerignore                                                            0.0s
=> => transferring context: 2B                                                                0.0s
=> [1/1] FROM docker.io/library/python:latest@sha256:9819e5616923079cc16af4a93d4be92c0c487c6e02fd902 35.4s
=> => resolve docker.io/library/python:latest@sha256:9819e5616923079cc16af4a93d4be92c0c487c6e02fd9027 0.0s
=> => sha256:07d1b5af933d2dfc3d0dd509d6e20534825e4a537f7b006a6cb5b8e5a1f20905 24.01MB / 24.01MB 4.4s
=> => sha256:9819e5616923079cc16af4a93d4be92c0c487c6e02fd9027220381f3e125d64a 10.04kB / 10.04kB 0.0s
=> => sha256:f285e872b4525229679829f02db72786ab1a5022d2ab55d7d67ac3cc0becd790 6.26kB / 6.26kB 0.0s
=> => sha256:1eb98adba0eb44a2e4facf9ca3626a4a66feedd0dd56d159cca90a35205744e7 64.40MB / 64.40MB 8.0s
=> => sha256:416855cb95a0dd2b7149c91f4af88549def1af9c6c4fa9ab78404578bc97271d 2.32kB / 2.32kB 0.0s
=> => sha256:23b7d26ef1d294256da0d70ce374277b9aab5ca683015073316005cb63d33849 48.49MB / 48.49MB 28.5s
=> => sha256:b617a119f8a27982374d94ec6eb3738ae3d38d6fc2c34c865813926cf596a621 211.33MB / 211.33MB 22.3s
=> => sha256:171e1bee194953b20a247468c6e03a4d1f67ba21a8034398f4ce3e620880f3ef 6.16MB / 6.16MB 9.5s
=> => sha256:e25cca11fd29c306a338d2d0397b465061f7a0c901916cc9ce7ecce3ed1e4823 27.37MB / 27.37MB 12.9s
=> => sha256:739b86d2a77825bc516695d8b66c3db5b776f53a426387a2ff7acf8567d07049 251B / 251B 13.1s
=> => extracting sha256:23b7d26ef1d294256da0d70ce374277b9aab5ca683015073316005cb63d33849 0.9s
=> => extracting sha256:07d1b5af933d2dfc3d0dd509d6e20534825e4a537f7b006a6cb5b8e5a1f20905 0.3s
=> => extracting sha256:1eb98adba0eb44a2e4facf9ca3626a4a66feedd0dd56d159cca90a35205744e7 1.2s
=> => extracting sha256:b617a119f8a27982374d94ec6eb3738ae3d38d6fc2c34c865813926cf596a621 3.4s
=> => extracting sha256:171e1bee194953b20a247468c6e03a4d1f67ba21a8034398f4ce3e620880f3ef 0.1s
=> => extracting sha256:e25cca11fd29c306a338d2d0397b465061f7a0c901916cc9ce7ecce3ed1e4823 0.4s
=> => extracting sha256:739b86d2a77825bc516695d8b66c3db5b776f53a426387a2ff7acf8567d07049 0.0s
=> exporting to image                                                                    0.0s
=> => exporting layers                                                                    0.0s
=> => writing image sha256:f285e872b4525229679829f02db72786ab1a5022d2ab55d7d67ac3cc0becd790 0.0s
=> => naming to docker.io/library/docker-python-1                                         0.0s
```



Выполним следующую команду, чтобы узнать размер итогового Docker image:

```
$ docker images --format "{{.Repository}}: {{.Size}}" |  
grep docker-python-1
```

```
ftlpp@user-IdeaPad-3-15IAU7:~/dockerfiles$ docker images --format "{{.Repository}}: {{.Size}}" | grep docker  
-python-1  
docker-python-1: 1.02GB
```

Проверим количество слоёв в итоговом образе:

```
$ docker image inspect docker-python-1
```

```
2,  
  "RootFS": {  
    "Type": "layers",  
    "Layers": [  
      "sha256:f7f2b929d8a55112adb1bc16fb8731045c9572b84d6dfbbdbd5dc6dd2bd9fe4",  
      "sha256:7f0053786e6e2411ce577e795ef6c278601de3faa0817c071801b11a3f83fd0d",  
      "sha256:b2bcd8ebb2be9869ca3198b4c521f18b612b4287ee4b705678d7b85383b71af",  
      "sha256:6c7c1b88da61a2ad4efd3cb93b6e31b0937d3f4121050a9d1c4ddd00485c7c68",  
      "sha256:7036bedc0c75773ae7704f8641e659f9ecb11863113505d4e72a014e78c077eb",  
      "sha256:1a6995d3d259ecbd29e3e4b528a508d33c0cf9da336e64f821ac3a9d4c70aa2b",  
      "sha256:637e74439babd2ce7b87c24a9389c753db135d374039f9ea2a7666a90ad859b8"  
    ]  
  },  
  "Metadata": {  
    "LastTagTime": "2025-04-14T13:18:11.487535199+03:00"  
  }  
}
```



Запускаем сборку образа на основе Dockerfile-python-alpine-2:

```
$ docker build . -f Dockerfile-python-alpine-2 -t docker-python-alpine-2
```

```
filipp@user-IdeaPad-3-15IAU7:~/dockerfiles$ docker build . -f Dockerfile-python-alpine-2 -t docker-python-alpine-2
[+] Building 15.1s (11/11) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile-python-alpine-2 0.0s
=> => transferring dockerfile: 218B 0.0s
=> [internal] load metadata for docker.io/library/alpine:latest 0.8s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [1/7] FROM docker.io/library/alpine:latest@sha256:a8560b36e8b8210634f77d9f7f9efd7ffa463e380b75e2e7 0.8s
=> => resolve docker.io/library/alpine:latest@sha256:a8560b36e8b8210634f77d9f7f9efd7ffa463e380b75e2e7 0.0s
=> => sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870 3.64MB / 3.64MB 0.6s
=> => sha256:a8560b36e8b8210634f77d9f7f9efd7ffa463e380b75e2e7a4ff4511df3ef88c 9.22kB / 9.22kB 0.0s
=> => sha256:1c4eef651f65e2f7daee7ee785882ac164b02b78fb74503052a26dc061c90474 1.02kB / 1.02kB 0.0s
=> => sha256:aded1e1a5b3705116fa0a92ba074a5e0b0031647d9c315983ccba2ee5428ec8b 581B / 581B 0.0s
=> => extracting sha256:f18232174bc91741fdf3da96d85011092101a032a93a388b79e99e69c2d5c870 0.1s
=> [2/7] RUN apk add --update-cache 1.0s
=> [3/7] RUN apk add python3 1.6s
=> [4/7] RUN apk add python3-dev 3.4s
=> [5/7] RUN apk add py-pip 1.1s
=> [6/7] RUN apk add build-base 5.5s
=> [7/7] RUN rm -rf /var/cache/apk/* 0.2s
=> exporting to image 0.6s
=> => exporting layers 0.6s
=> => writing image sha256:c3346f9ecdec1bd2432ee3cc8f9f03eb374259ca6cb377920b59f75c572f7ff4 0.0s
=> => naming to docker.io/library/docker-python-alpine-2 0.0s
```

Выполним следующую команду, чтобы узнать размер итогового Docker image:

```
$ docker images --format "{{.Repository}}: {{.Size}}" | grep docker-python-alpine-2
```

```
filipp@user-IdeaPad-3-15IAU7:~/dockerfiles$ docker images --format "{{.Repository}}: {{.Size}}" | grep docker-python-alpine-2
docker-python-alpine-2: 412MB
```



Проверим количество слоёв в итоговом образе:

```
$ docker image inspect docker-python-alpine-2
```

```
"RootFS": {
  "Type": "layers",
  "Layers": [
    "sha256:08000c18d16dadf9553d747a58cf44023423a9ab010aab96cf263d2216b8b350",
    "sha256:b7f11cb845215a5b2c0469233faa7f60184e34e112996d93629d3eeb48204836",
    "sha256:75f392078d547e16c90e11274e13a7e99b64765d05e6509d1fde686046fdbbbd",
    "sha256:219a66cf3bd27656abf4c0fd01f674a118d2d83deaaf9ee51dbec9f111cb7a69",
    "sha256:46003273adb3c4727b176923279394837358365c735f000c0fca3f872e65f571",
    "sha256:1664a719d28f8e7c8cfb26ee46857dd6ee1fdc72f7245a40237ce8dd97fccaf0",
    "sha256:a3c741a141172a1d79749d3fe4716735a0876b9d85a51b83156dcc723d974a5e"
  ]
},
"Metadata": {
  "LastTagTime": "2025-04-14T13:19:38.720563534+03:00"
}
```

Нам удалось достичь сокращения размера Docker image за счет использования более легковесного базового образа, но не путем сокращения слоев.

Теперь запускаем сборку образа на основе Dockerfile-python-alpine-3:

```
$ docker build . -f Dockerfile-python-alpine-3 -t docker-python-alpine-3
```

```
filipp@user-IdeaPad-3-15IAU7:~/dockerfiles$ docker build . -f Dockerfile-python-alpine-3 -t docker-python-alpine-3
[+] Building 20.5s (6/6) FINISHED
=> [internal] load build definition from Dockerfile-python-alpine-3                                docker:default 0.0s
=> => transferring dockerfile: 187B                                                                0.0s
=> [internal] load metadata for docker.io/library/alpine:latest                                  2.2s
=> [internal] load .dockerignore                                                                    0.0s
=> => transferring context: 2B                                                                      0.0s
=> CACHED [1/2] FROM docker.io/library/alpine:latest@sha256:a8560b36e8b8210634f77d9f7f9efd7ffa463e380 0.0s
=> [2/2] RUN apk add --update-cache python3 python3-dev py-pip build-base && rm -rf /var/         17.0s
=> exporting to image                                                                                1.2s
=> => exporting layers                                                                                1.2s
=> => writing image sha256:7ba9788432ed459e6e5a140dc182ef37f0422f75be521a147d6f55e779f268d3      0.0s
=> => naming to docker.io/library/docker-python-alpine-3                                           0.0s
```



Выполним следующую команду, чтобы узнать размер итогового Docker image:

```
$ docker images --format "{{.Repository}}: {{.Size}}" |  
grep docker-python-alpine-3
```

```
filipp@user-IdeaPad-3-15IAU7:~/dockerfiles$ docker images --format "{{.Repository}}: {{.Size}}" | grep docker  
-python-alpine-3  
docker-python-alpine-3: 409MB
```

Проверим количество слоёв в итоговом образе:

```
$ docker image inspect docker-python-alpine-3
```

```
{  
  "RootFS": {  
    "Type": "layers",  
    "Layers": [  
      "sha256:08000c18d16dadf9553d747a58cf44023423a9ab010aab96cf263d2216b8b350",  
      "sha256:002421c7fd7f23eb71a6e342eca3cacef1ef677115a7312c764675b33399b0c0"  
    ]  
  },  
  "Metadata": {  
    "LastTagTime": "2025-04-14T13:21:26.715512059+03:00"  
  }  
}
```

Теперь мы оптимизировали и размер Docker image, и количество его слоёв. Данный образ почти в 3 раза легче и быстрее, чем первоначальный